GEPM 463

PROJECT MANAGEMENT & FINANCE

Welcome!

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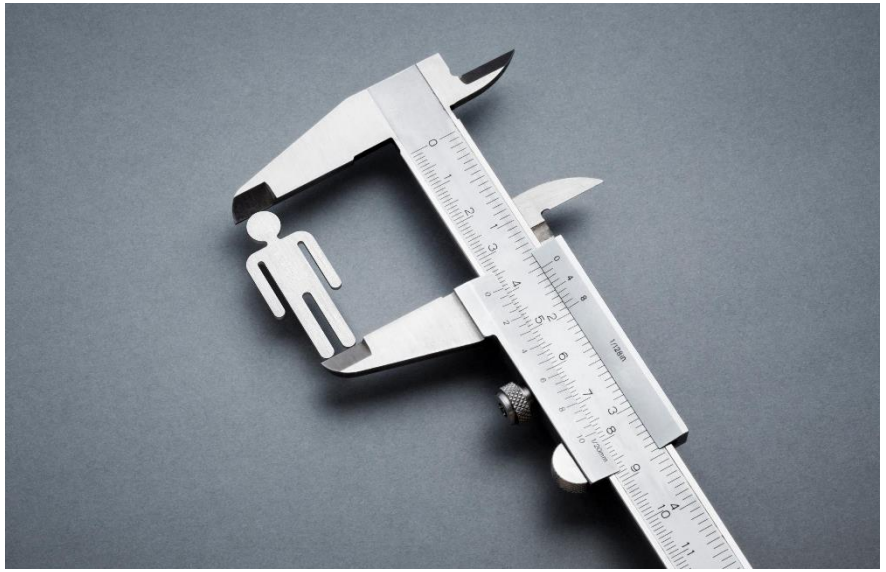
Let's Get Started

- Your Turn…..
- Tell us about you
  - Your Name
  - What does Project Management mean to you?
  - Why is Project Management important?



About This Course

- Google Classroom
  - Your Code: fl3a434

- Your Course-related information will be posted here.
  - You can access the information at any time.
- You can communicate and post your questions.
  - Allow 24 hrs. for reply.
- Weekly Lectures can be accessed.
- You can access your eBook.

Your Responsibility on a Weekly Basis

- Check your Google Classroom 2-3 times/week.
- Download your Weekly Lecture Notes before your class.
- Check for and review your Instructor's Announcement (if any).
- Take actions!

- Read the Lecture Notes.
  - Ask Questions.
- Review your class notes.
  - Ask questions to further clarify.
  - If you still don't understand, read the text Chapter.
  - All your questions will be from the Lecture Notes!!

What will be covered in this course

- This course has been designed to understand the overlapping connection between engineering and management with financial matters in an organization through the study of varied management practices and finance as an engineer.

What will be covered in this course

- **Objectives**

    1. To introduce some aspects of business management and business organization.
    2. To identify the tools and techniques needed to lead any project to its intended conclusion.
    3. To introduce sales fundamentals include understanding the customer and the competition, sales strategy, sales management, product positioning, product life cycle, sales structures, margins, and prospecting for new customers.

What will be covered in this course

Course Content

- Management Functions and Organization: Evolution; Management functions: organization, theory and structure, span of control, authority delegation, manpower planning.
- Personnel Management: Importance, need hierarchy, motivation, leadership, wage incentives, performance appraisal, participative management;
- Operation Management: Production planning and control (PPC) functions, quantitative methods applied in production, quality management, location and layout planning, safety and loss management.
- Cost management elements of cost of products, cost centers, and allocation of overhead costs; Management accounting : marginal costing, standard costing, cost planning and control, budget and budgetary control;
- Development and planning process; annual development plan; National budget, Financial management : objectives, strategy, financing, performance analysis of enterprise, investment appraisal, investment criteria; Management Accounting : Cost planning and control, budget and budgetary control.
- Marketing Management: Concepts, strategy, sales promotion, patent laws. Technology Management; Management of innovation and changes, technology life cycle, Case studies.

Marks Distribution at a glance

- Total Available Points:

- Class Tests:
- Midterm Exam:
- Final Exam:
- Group Project/Research Paper:
- Final Presentation:

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What is a Project?

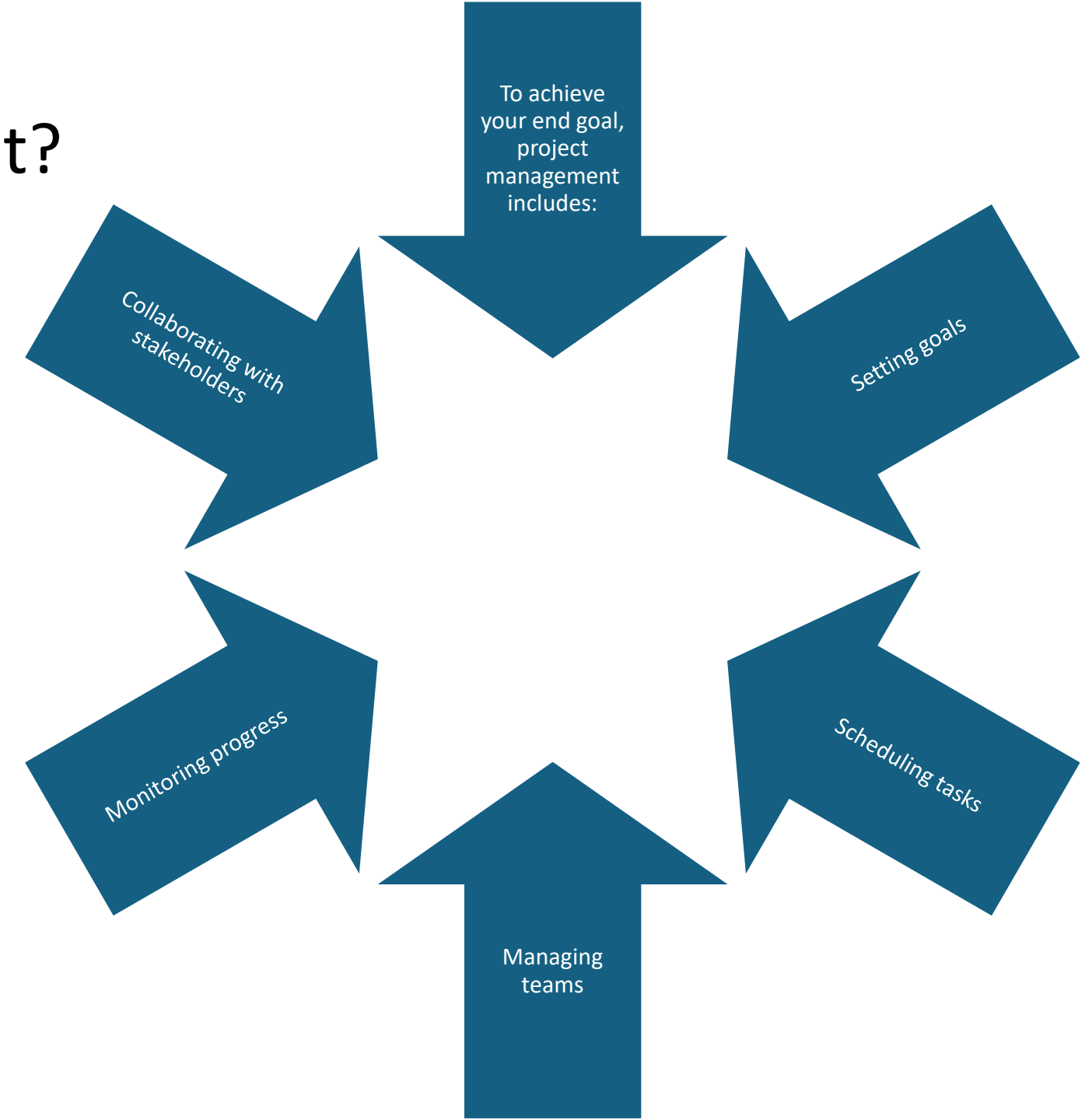
- A **project** is “a temporary endeavor undertaken to create a unique product, service, or result” (PMBOK® Guide, Fifth Edition, 2012)
  - Projects end when their objectives have been reached or the project has been terminated
  - Projects can be large or small and take a short or long time to complete

Examples of IT Projects”

- A team of students creates a smartphone application and sells it online
- A company develops a driverless car
- A small software development team adds a new feature to an internal software application for the finance department
- A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus

What is Project Management?

- Project management (PM) is the practice of planning and managing projects, from start to finish.
  - It involves the coordination of all the activities and resources you need to meet deadlines and achieve your desired project outcome.
  - For example, your goal might be to launch a new product, develop software, or plan a networking event.



5 Phases of Project Management

- Initiating
- Planning
- Executing
- Monitoring/Controlling
- Closing

Initiating

- As with any good project, preparation is vital to complete the work on time and within budget.
- During the project initiation phase, determine the following to create the project foundation:

- **Project scope**: This covers the specific limits and boundaries for the project, including what you'll accomplish and what you will *not (*i.e., the project objectives*)*. Establishing these boundaries prevents something called "scope creep" where the deliverables change and your project deadlines may get pushed back.
- **High-level project overview:** This outlines the resources, time, and goals required to complete a project and a strategy to monitor these requirements over time.
- **Budgets**: You'll liaise with stakeholders to determine how much money is available for your project, and the cost required to achieve a successful project.

Planning

- Drawing up a clear and intentional project plan ensures all team members move toward the same goal.

During the project planning phase, you'll:

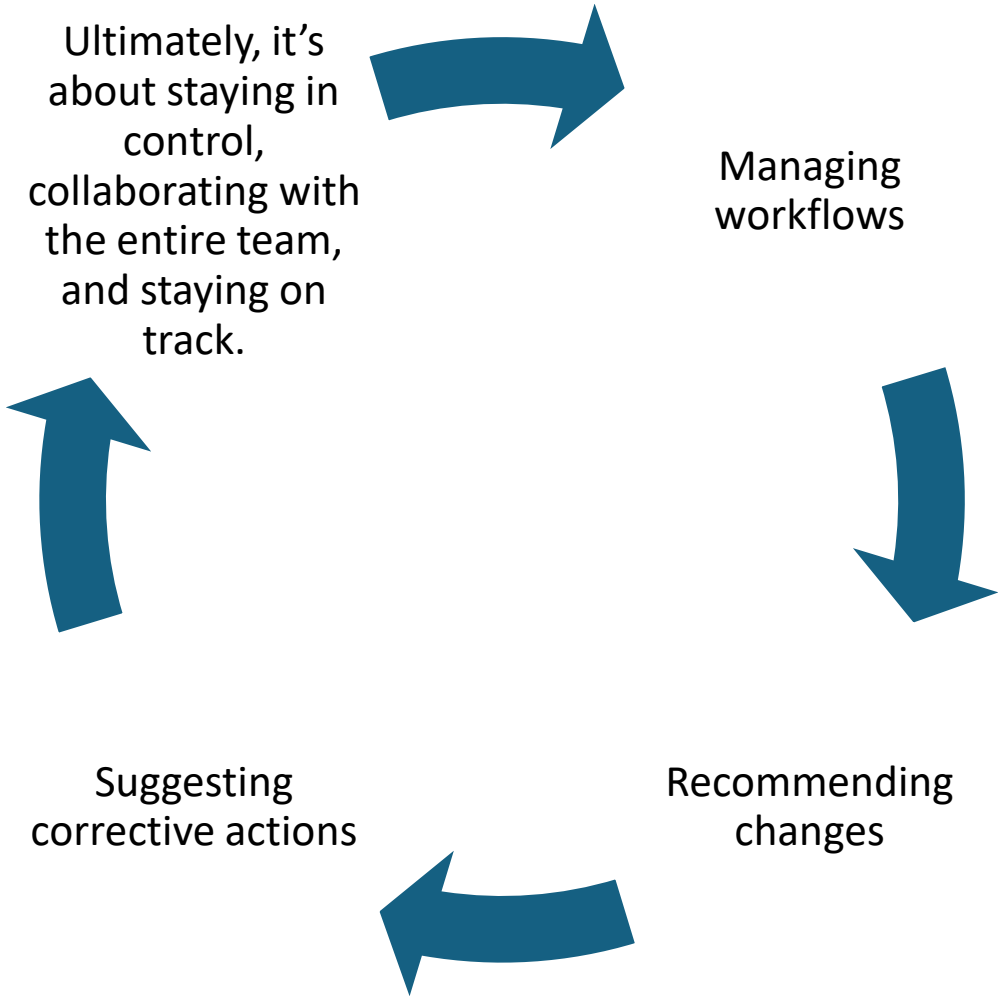
- Select your project team members
- Outline your deliverables
- Estimate your required project resources
- Determine any associated activities
- Set key milestones and dates, including the final project completion date

Planning – cont'd

- During this process, you can also outline the team's project management methodology.
- There are many to choose from, including the below popular options. We'll get into more specifics around some of these later in this project management guide.

Agile	Waterfall	PRINCE2
PMBOK	Scrum	Lean
	Kanban	

Executing



- During the project execution phase, you will carry out the details of your project plan (or project deliverables) to deliver your products to your specific stakeholders.
- This stage occurs simultaneously with the monitoring and controlling phase (below) and might involve:

Monitoring/ Controlling

- Constant monitoring is essential in any project life cycle as it helps project managers answer the question:
  - where *are* we? vs.
    where *should we* be? according to the project plan.

Effective monitoring requires:

- Regular, consistent project "check-ins"
- Use of proper project documentation and tracking tools to visualize project progress (such as Gantt Charts,)

Closing

Concluding any formal contracts or agreements

A full review or audit of what went well, what didn't go as planned, and how future teams could learn from this project

This final step, sometimes known as "project delivery" or the project closeout phase, is when you wrap up all activities and deliver the final product.

This handover might be to the client, an internal team or an external stakeholder. Some additional components could be:

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Lecture 2

Introduction

- With the Best engineering skills and knowledge, professionals cannot just sustain since any operational issues require applying management techniques.
- Any engineer by profession is also a manager, as engineers also qualify in the classical test of managerial roles such as interpersonal roles, informational roles, and decisional roles.

Management – Definition!

- Operationally –

"A dynamic process, that helps to get things done, through and with the efforts of people"

- "Management is the exercise of drawing the straight line from what you are and where you are; to what you want to be; where you want to go.
- The straight line connotes the economy in time and means – materials, and human; a journey starting from the starting point to the destination."

"Management is the optimization of constraining resources to achieve some intended goals.

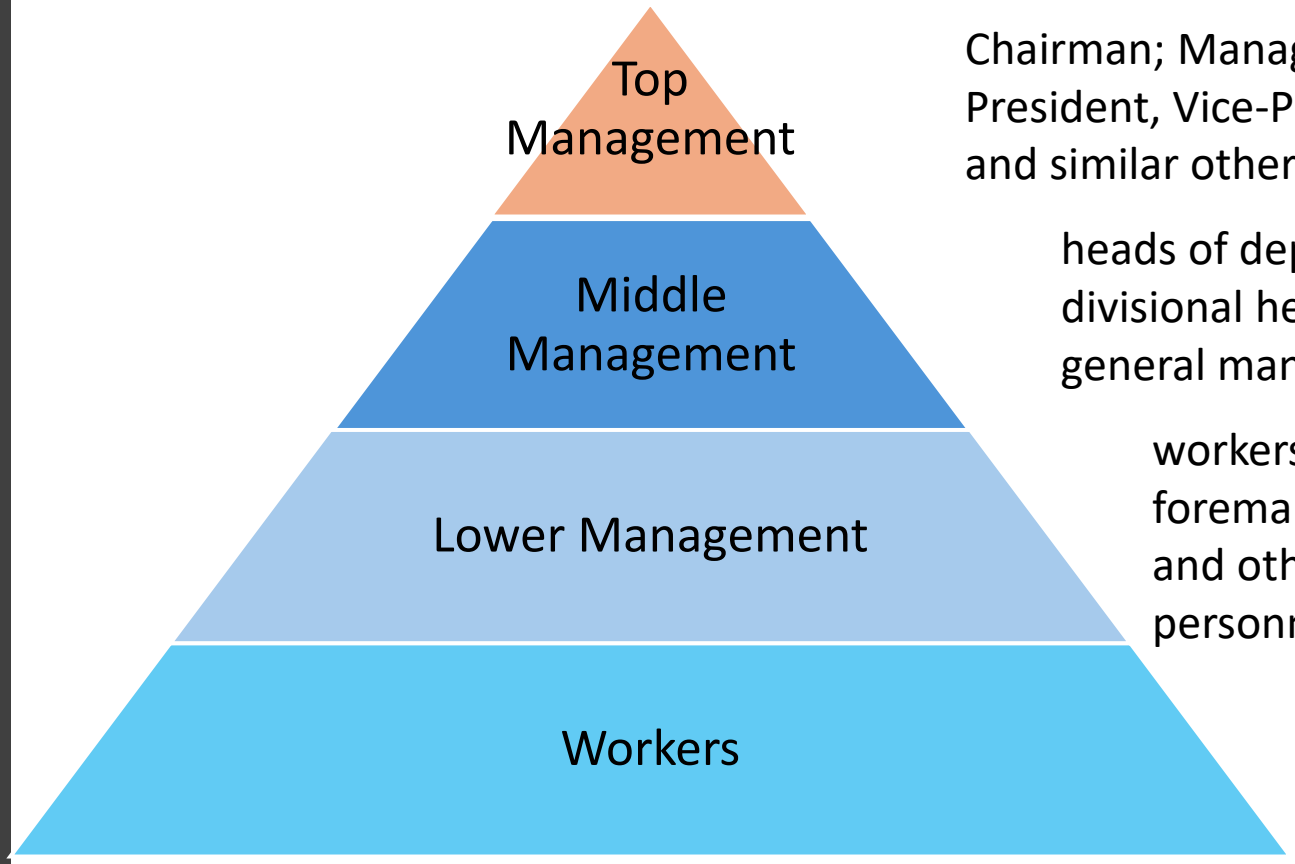
- "Resources" is a broad term and it encompasses everything that we require as inputs (including knowledge and information)
- Resources are not available in abundance, thus resource constraints exist.

Nine Elements of Management

In which Management is conceived, evolved, and implemented.

1. **Horizon** – large or small, it has been defined for management within which they must perform
2. Must be an **Organization**, which gives body to the management
3. **Universal need** for planning, implementation, and monitoring
4. Any management must be properly equipped with its **staff**
5. Management needs **Leadership and Direction**
6. **Communication** is the blood of the management process
7. There is a need for all-pervasive **Coordination**
8. Requires constant **Evaluation, monitoring, and control**
9. **Innovation,** pride of place in the competitive management process

Types of Management: Levels

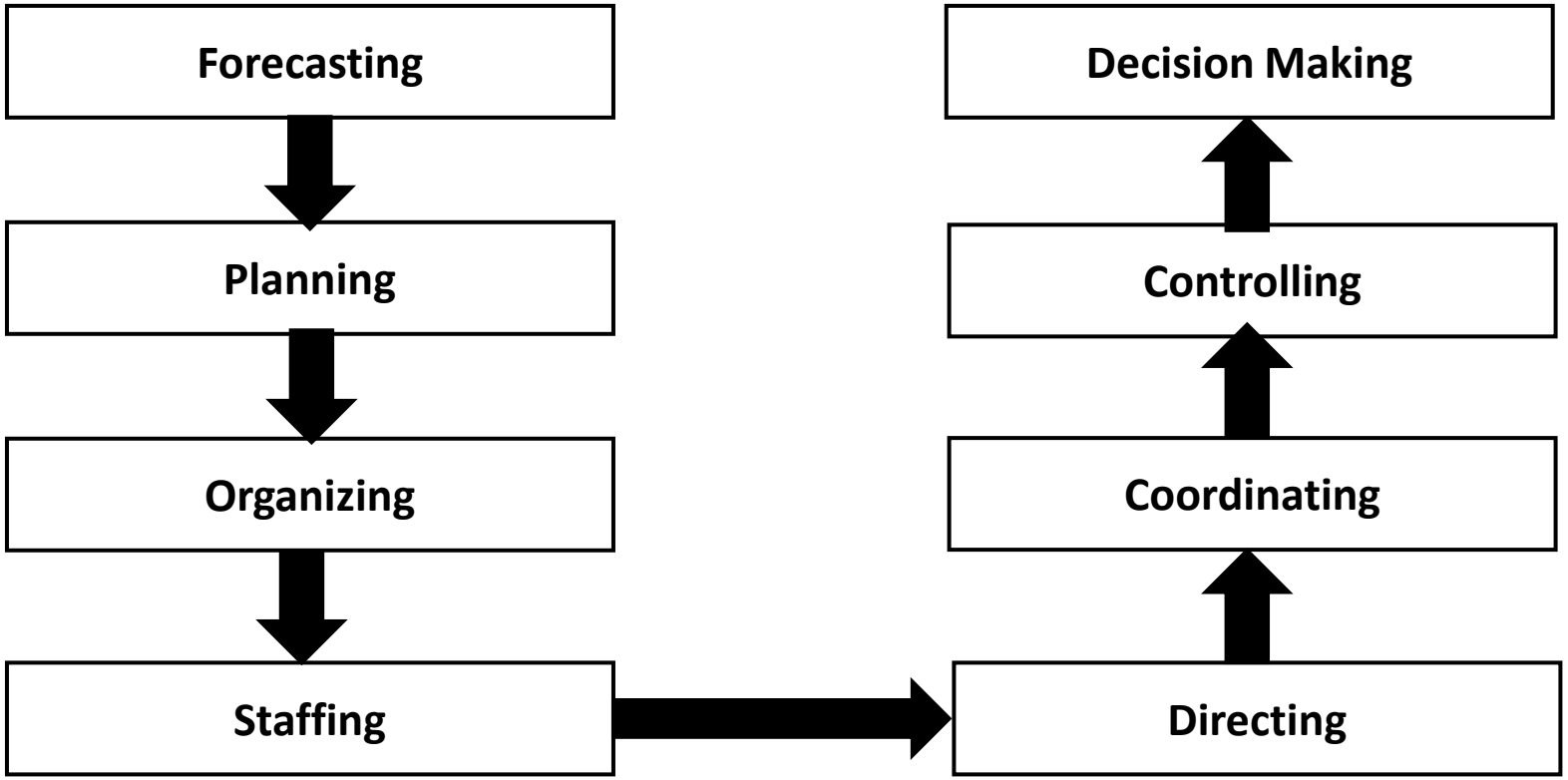


Chairman; Managing Director, President, Vice-President; and similar other positions

heads of departments, divisional heads, general managers, etc

workers like supervisor, foreman, superintendent and other operating personnel

Process of Management



The management process, involving four managerial functions, is an integration of the manager's functions.

Planning

(Key Functions of Management)

The planning function of a manager includes activities that lead to the definition of ends and the management of appropriate means to achieve the defined end.

Planning involves advance decisions related to what, when, why, how, and who

- Self-appraisal to determine the current position.
- Study of the environment around the organization.
- Specifications of goals and objectives and means to achieve them.
- Framework of policies, procedures, standards, and anticipated course of actions.
- Forecasting.
- Resources to achieve the forecast.
- Revision of plans and adjustments in changed situations.
- Coordination of processes involving planning.

Organizing

(Key Functions of Management)

This framework includes people, tasks, resources, and performance with respect to the organizational goals.

Organizing involves those activities of the management that are performed to translate the required activities of plans into a structure of task, authority, and responsibility.

- Defining the nature and content of each job in the organization.
- Setting the base for grouping the jobs together.
- Deciding the size of groups.
- Delegating authority to assigned managers

Directing & Motivating

(Key Functions of Management)

Directing and motivating channelize the organizational behavior for attainment of corporate goals.

Directing and motivating stimulate the organization to undertake actions along the plan.

- Communication and explanation of objectives to the subordinates.
- Assign the performance standards.
- Helping subordinates through proper guidance and personal interaction to meet the standards of performance.
- Reward for better performance both financially and written appreciation.
- Praise and censuring the employees wherever required.
- Management of change through proper communication and building confidence among different layers of management and workers.
- Coordination in the entire process

Controlling

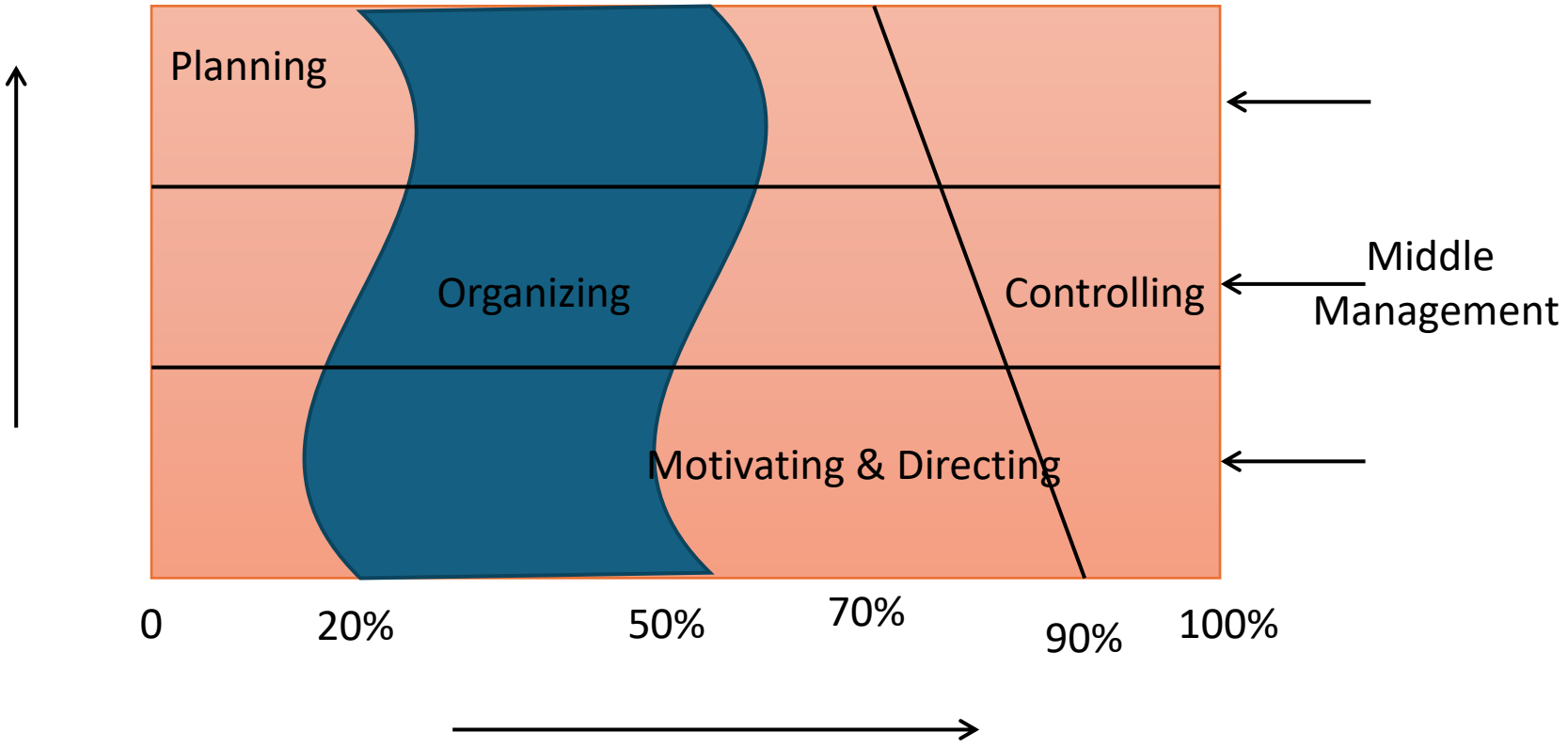
(Key Functions of Management)

The control function is useful for (a) short-range corrections through better directing, (b) moderate corrections through re-engineering benchmarking, and (c) long-term corrections through re-defining goals and new planning endeavors.

Controlling relates to measuring the performance against the goal, determining the causes of deviations from goals, and taking corrective actions for improvement. Managers are required to assure that the actual outcomes are consistent with the planned outcomes.

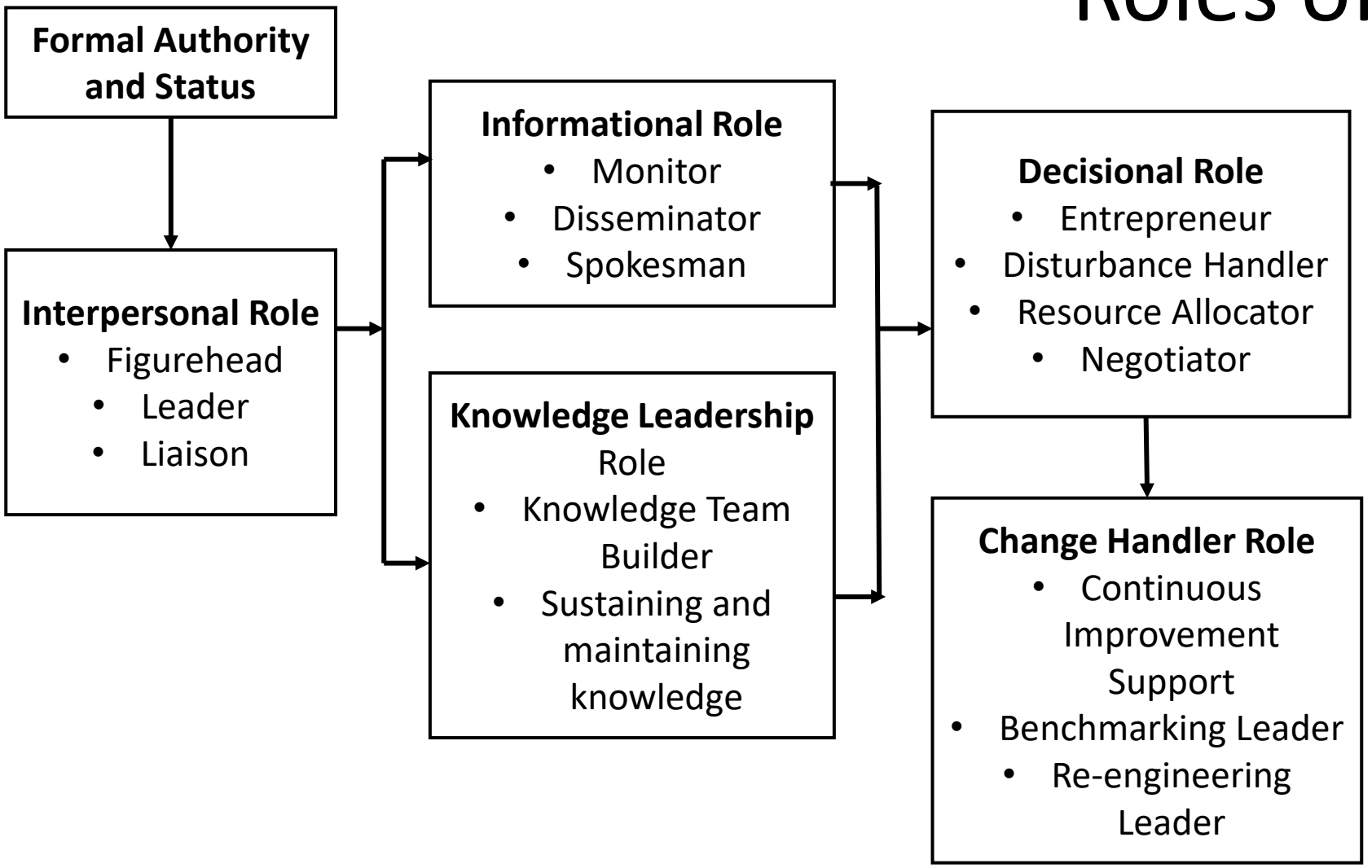
- Standard of performance.
- Information related to gaps in standard and attained performance.
- Corrective actions to bridge the identified gaps.

Time spent (%) by each level of management



Percentage of time spent in particular function of management

Roles of Manager



Modified and enlarged in scope from: Henry Mintzberg, 1973,
The nature of Managerial Work, Harper & Row Pub NY, pp 93-94

Managerial Role

INTERPERSONAL

- **Figure Head**
  - Representing the organization to perform ceremonial duties. It means symbolic head.
- **Leader**
  - Influencing subordinates to achieve their goals and objectives (motivating and activating).
- **Liaison**
  - It means maintaining a self-developed network of outside contacts and information.

Managerial Role

INFORMATIONAL

- **Monitor**
  - Collecting information concerning the organization and shortlisting relevant information (thorough understanding of the organization and environment).
- **Disseminator**
  - Sharing relevant outside information with subordinates (filtering, clarifying, interpreting, and integrating different information so that value-added knowledge emerges for the organization)
- **Spokesperson**
  - Maintaining protocol to share information with outsiders (an expert to clarify the organizational plan, policies, actions, and results).

Managerial Role

DECISIONAL

- **Entrepreneur**
  - Focusing on innovation and change within the organization (searching the organization and its environment for opportunities and initiating the “improvement (or change) process” to bring about transformation)
- **Disturbance Handler**
  - This means responsibility related to corrective actions when the organization faces sudden, unexpected disturbances. His actions include review and rectification of the crisis
- **Negotiator**
  - This means representing the organization at major negotiations. His actions include bringing advantage to the organization during the process of negotiation
- **Resource Allocator**
  - Optimizing resource allocation to different competing needs within the organization. actions involve scheduling, budgeting, allocation of duties to subordinates, authorization, etc

Managerial Role – Knowledge Leadership Role

- **Knowledge Team Builder**
  - This means that the Managers should create teams that have expertise in certain areas
  - This is done through regular updating of knowledge through seminars, journals, internet research, and the adoption of technology.
  - Activities include finding the right people, who can share the same expertise in building knowledge-base.

- **Sustaining and Maintaining Knowledge**
  - This is related to knowledge management.
  - Activities include documenting and sharing the expertise among group members.
  - If an individual leaves the organization, the knowledge should stay with other members of the group.

Managerial Role – Knowledge Leadership Role

- Continuous Improvement Supporter
  - Managers' role is to develop a quality culture and teambuilding.
  - Problems are identified and solved for small but gradual improvement
  - The role of the manager is to tell everybody that there exists a better way of doing the thing which we are doing now. It is the path that TQM also advocates.

- Benchmarking Leader
  - Involves identifying "best practices" or world-class performers in your area and identification of gaps.
  - This gap is bridged through systematic planning and leadership.
  - Manager's actions involve the identification of benchmarks, building teams to make changes, and evaluation of performance during the change.

- Re-engineering Leader
  - Is the total, radical redesign of the system. Managers have to prepare resources (including subordinates) for a total transformation.
  - Unlike continuous improvement, which is gradual, and benchmarking, which is moderate, re-engineering is a dramatic transformation. So, the risk is higher.
  - Therefore, its management is more difficult as compared to the other two change process.

Conclusion

Now that we understand “Management”,
let’s look at “Organization”.

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Organization Planning, Design, and Development

Lecture 3

Organization - Introduction

Organizations are formed – in pursuit of an objective call for the joint efforts of two or more individuals.

- **Major Components:**
- Composed of individuals and groups of people
- Seek the achievement of shared objectives through the division of labor
- Integrated by information-bound decision processes to coordinate the resulting tasks

Orgaization – Definition!

“It is the **Grouping of activities** necessary to attain organizational objectives and assignment of each grouping to a manager with authority necessary to supervise it.” – (Koontz and O’Donnell)

“Is the **Process of identifying and grouping work** to be performed, defining and delegating responsibility and authority, and establishing relationships to enable people to work most effectively together in accomplishing objectives.” – (Allen)

An Organization Has…

i. A system
ii. Well-established structure
iii. People to work and deal with each other in a coordinated and cooperative manner
iv. Grouping of work
v. Established relationship for authority and delegation
vi. Attainment of common goals
vii. Internal structure for performance
viii. Defined roles and responsibility of each person
ix. A constituent of (a) division of labor, (b) identification of the source of authority, and (c) establishment of enterprise relationship.

Organizational Planning

Purpose of Organizational Planning:
Organization planning defines or reshapes the organization structure for:

a. Clarifying Objectives, roles, and relationships;
b. Determining the management resources required now and in the future; and
c. Providing information on the job requirements
    i. right people can be appointed
    ii. Adequate training can be given
    iii. Payments to the staff are commensurate with their relative levels of responsibility and value to the organization.

The organization structure is the framework for carrying out the task of management. The overall task has to be divided into a variety of activities, and means have to be established for the direction, coordination, and control of these activities.

Stages followed in an Organization Planning (Organization Planning Process)

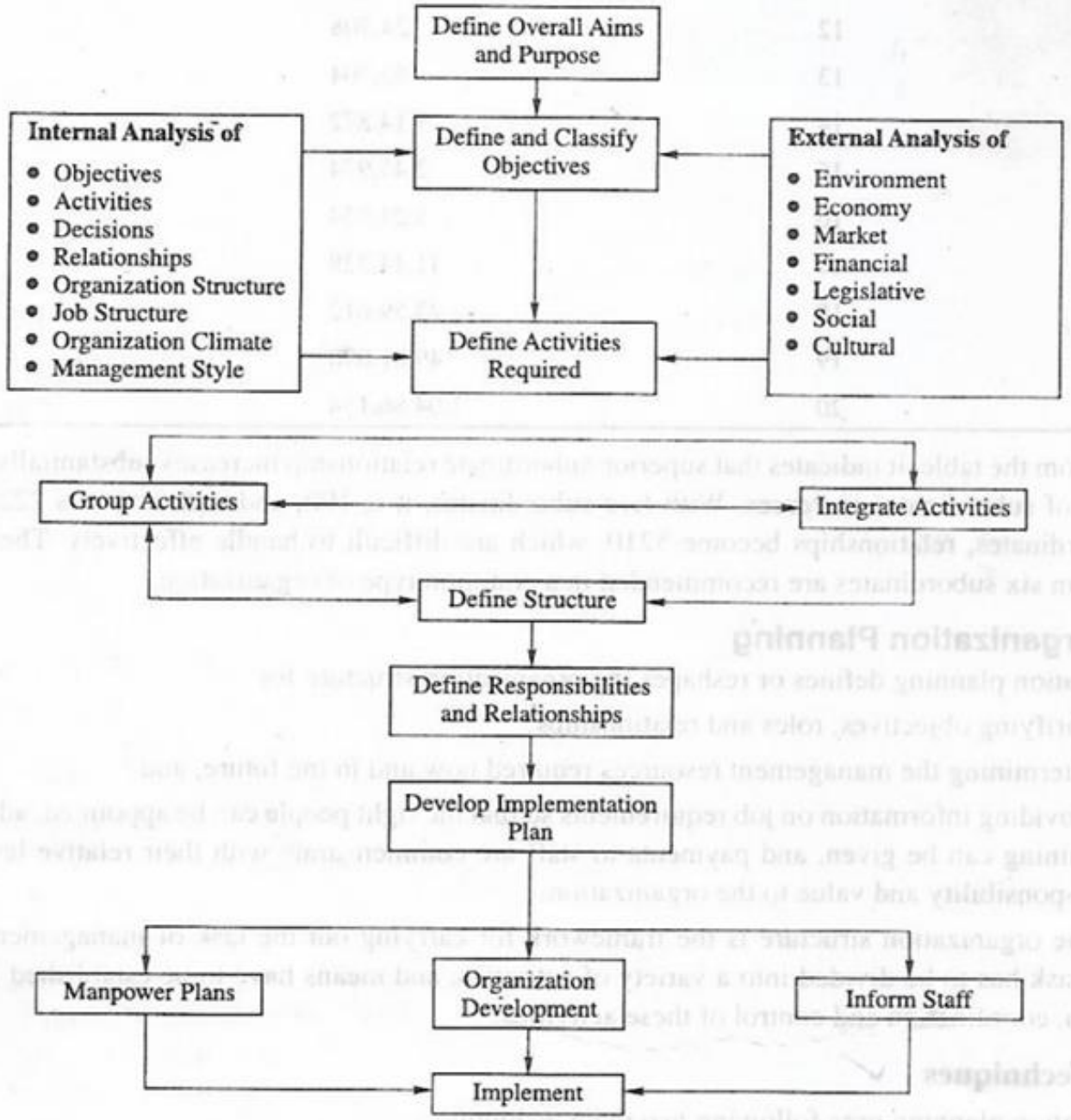


Fig. 2.1 Organization planning programme

Organizational Analysis Examines

1. **Objectives**: To find out what they are and how clearly they are defined and understood at all levels.
2. **Activities**: To establish what work is done and what work needs to be done by the organization to achieve its objectives.
3. **Decisions**: To study where and by whom the key decisions are made and how work is delegated and decentralized.
4. **Relationships**: To define what interactions and communications take place between people in the organization so that an assessment can be made of the extent to which the grouping of activities, lines of communication and information systems facilitate effective management and coordination.
5. **Organization Structure**: To find out
   - How activities are grouped together
   - The spans of control of senior and middle managers and
   - The number of levels in the management hierarchy

6. **Job Structure**: To determine the content of individual jobs in terms of duties, responsibilities, and authorities. The aim is to establish the degree to which tasks are logically and clearly allocated and to indicate whether job holders are clear about what they are expected to achieve.

7. **Organization Climate**: To get a feel of the working atmosphere with regards to teamwork and cooperation, commitment, communication, creativity, conflict resolution, participation, and confidence and trust between people.

8. **Management Style**: To find out what sort of approach to management is being practiced, especially at the top.

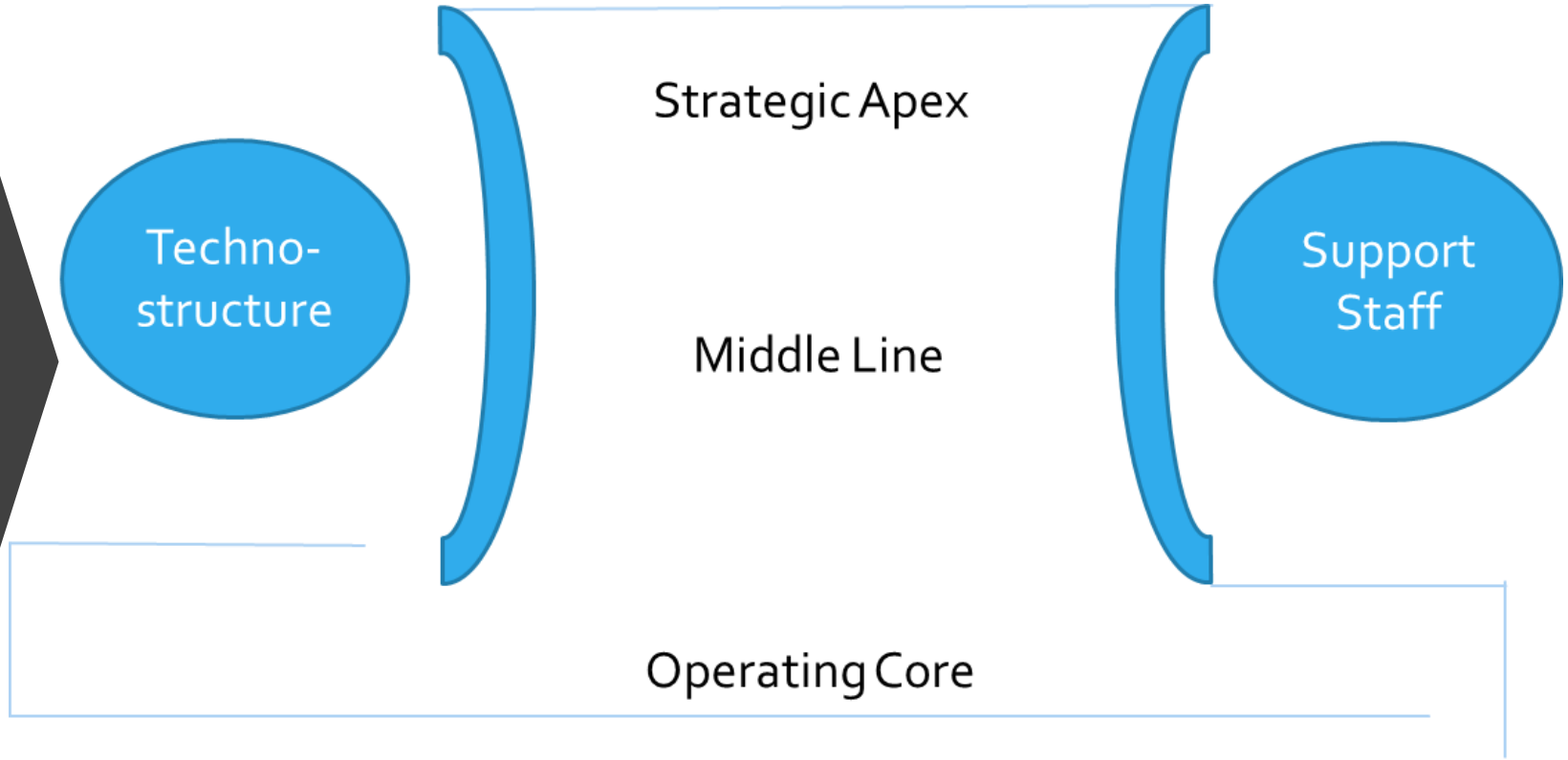
9. **Management:** To establish

- The extent to which the existing organization has been built around the personalities, strengths, and weaknesses of the key people in it and
- The availability of the quality of people required to facilitate necessary changes in the organization structure.

Symptoms of an Inadequate Organizational Structure

1. Lack of opportunities for development
2. Insufficient time for development
3. Intensive antagonistic environment
4. Lack of definition in portfolio business planning
5. Lack of coordination with the sister division
6. Excessive duplication of functions in different units of the firm
7. Excessive dispersion of functions in one unit of the firm
8. Poor profit performance and low return expectations

Five Basic Parts of an Organization



Five Basic Parts of an Organization

Mintzberg conceptually describes an organization as typically having Five basic parts: Fig. 2.3

1. **Operating Core**: This part consists of those who perform the basic work relating to production or services.
2. **Strategic Apex:** Comprises people having overall responsibility for the organization, namely the chief executive and other top-level managers.
3. **Middle Line**: It is the linking pin between the strategic apex and the operating core. The chain runs from the senior managers down to the first-line managers.
4. **Technostructure**: This is made up of analysts whose job is to control, stabilize, and standardize patterns of activity in the organization.
5. **Support Staff**: Support staff is engaged in large organizations to encompass more and more boundary activities. Can be found at various levels of hierarchy.

Organizational Development (OD)

Definition:

Is concerned with the planning and implementation of programs designed to improve the effectiveness with which an organization functions and responds to change.

Outcomes of OD:

- More participative management.
- Integration of individual goals with organizational goals.
- OD is intended to create an internal environment of openness, trust, mutual confidence, and collaboration.

Objectives

- Improve Organizational Performance as measured by profitability, market share, and innovativeness.
- Better Adaptability to its environment.
- Willingness of Members to face organizational problems and find creative solutions.
- Improve Internal behavioral patterns

Models of OD (Change Process – Behavior)

There are 3 popular Models of OD:

1. Kurt Lewin's Model

2. Larry Greiner's Sequential Process Model

3. HJ Leavitt's System Model

Kurt Lewin's Models of OD

Kurt Lewin's Model – It can be divided into 3 stages:

-- **Stage 1**: Unfreezing (Creating the need for change, motivating the people for change, and minimizing resistance to change)

-- **Stage 2**: Changing (Transition from old behavior to experimentation with new behavior in terms of cognitive redefinition through identification and scanning.

-- **Stage 3**: Re-freezing (Stablishing and integrating the change by reinforcing the new behaviors and integrating them into one's personality as well as in formal and interpersonal relationships.

Characteristics of a Successful OD Programme

1. It is a planned program that involves the whole of an organization or a relatively autonomous unit within an organization.
2. It is designed to create organizational conditions which will directly help the institution to accomplish a specific strategy.
3. The senior management is personally committed to the goals of the program.
4. It is a long-term effort (takes 2-3 years)
5. The program is action-oriented (the program is designed from outside to generate action by those involved)
6. Particular emphasis is placed on changing perceptions, attitudes, and behavior, not solely on the structural or procedural change
7. It relies on some form of experience-based learning through which participants can examine the present situation, define new goals, and explore new ways of achieving them.
8. The basis of all programs is the groups or teams from which the organization is constructed.

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Management Planning & Control

Lecture 4

Planning

Planning is a process of visualizing intelligently appraising the future needs and purposes, identification and evaluating the opportunities and risks.

Planning is a process:

- It is a dynamic and continuous activity
- Protect changing conditions both from within and outside
- Management has no control over the economic force without it
- Plans can be effectively achieved in most organizations with good control
- Planning is always a prerequisite for controlling
- Planning seeks to set goals and programs and
- Control seeks to secure performance in accordance with plans.

Management Planning

- **Analyzing Factors Affecting the future**
- The planner should start looking into the conditions which affect the organization's future opportunities.
- These conditions are (a) environment, (b) internal conditions, and (c) obligations of the organization.
- In essence, an organization must be prepared and able to change the course of its business to meet customer demands and to find improved ways of service through better products, quality, and value.

The following ten elements are included to some degree or other in any management planning activity:

- Organization Planning
- Product Planning
- Facility Planning
- Operations Planning
- Material Supply and Inventory Planning
- Financial Planning
- Commercial Planning
- Production Planning
- Human Resource Planning
- Development Planning

Modern Management Principles

- Management is the process of getting things done by other people
- Management as an activity is universal but does not take the same form in all situations
- Organization is complex as well as an open system
- Organization is composed of technical and sub-systems
- Appropriate and timely Innovation is needed for sustainable survival
- No single best way to organize a business
- Knowledge (product), process (know-how), or knowledge of the market – is the essence of competition
- Management is a process involving a mix of rational, logical decision-making and problem-solving activities and intuitive, judgemental activities.

General Management

General management is a practice that plans, organizes, directs, and controls a number of interrelated operations and supporting services in order to achieve defined objectives.

It is a discipline requiring the use of a number of managerial skills rather than a technique in itself. However, the skills of general management and the effectiveness with which the activities are directed and controlled to achieve the desired result are dependent on the understanding and use of a wide range of management techniques

Four Functions of General Management

- **Planning:** Deciding where the organization should be going and how it should get there. This requires the appraisal of external and internal changes and constraints, forecasting, setting objectives, developing strategies and policies, and preparing action plans.

- **Organizing:** Deciding who does what. This requires the definition and grouping of activities, defining responsibilities, and establishing means of communication, coordination, and control.

- **Directing**: Ensuring that people know what to do and when to do it, and exercising leadership to get individuals to work to the best of their ability as part of a team.

- **Controlling**: Measuring and monitoring results, comparing results with plans, and taking corrective action when required

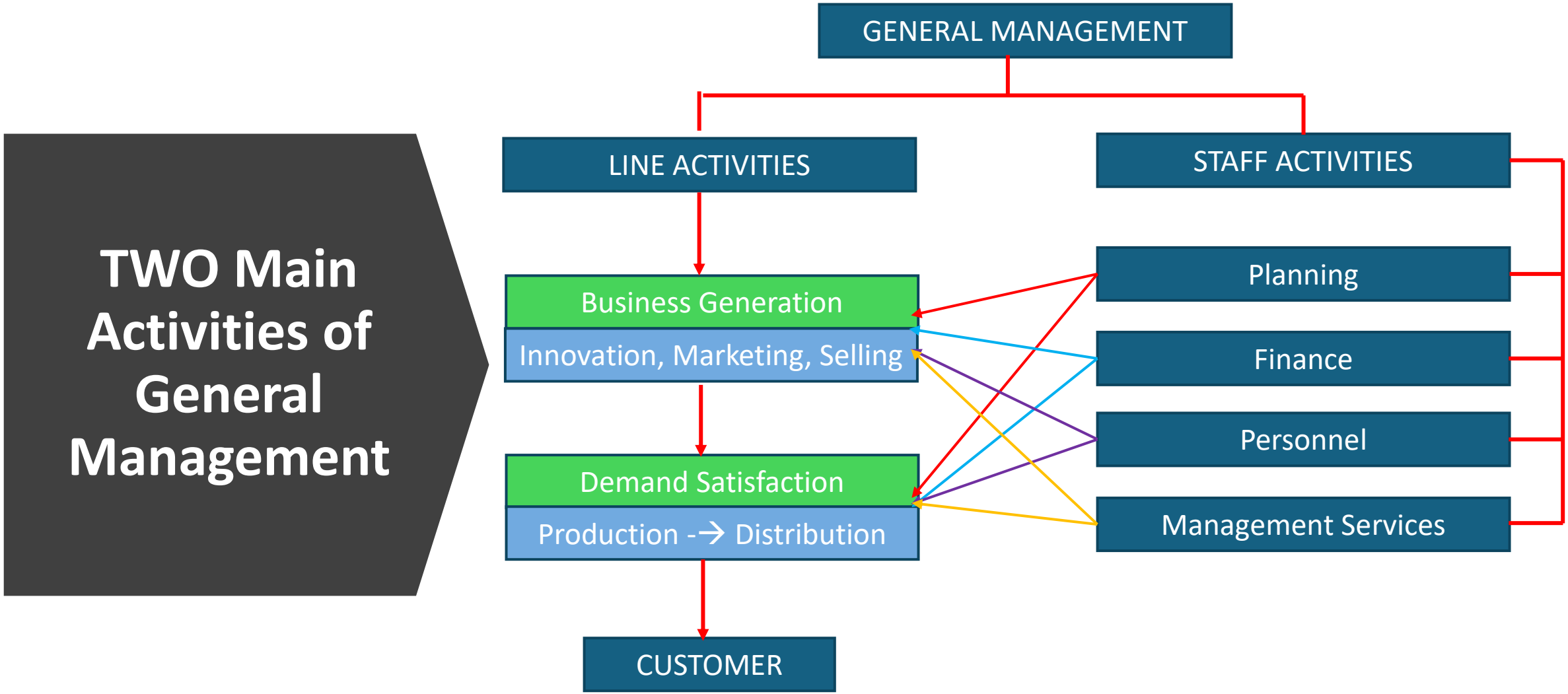


Figure. 3.2 of page 32 - Line Activities vs. Staff/Service Activities

TWO Main Activities of General Management

Line Activities

In a commercial enterprise, these comprise:

- **Business generation**: Innovation (new product development), marketing, and selling.
- **Demand satisfaction**: Manufacturing the product to meet demand and distributing it to the customer. Feedback from the customer in the form of new orders and reactions to the product will affect the business generation activities

Staff or Service Activities

These ensure fulfillment of line activities by means such as:

- Giving a sense of direction through planning and budgeting.
- Providing finance and the means of planning and controlling expenditure. - - Providing manpower in the quantities and qualities required.
- Providing management services and management support activities which include information technology, operations research, work-study, and other activities designed to improve the efficiency and effectiveness of the organization.

Management Control

Management control is the process of assuring the efficient accomplishment of set objectives of an enterprise.

In the turbulent environment faced by management, control is necessary to:

(1) anticipate problems,

(2) measure performance against standards,

(3) take corrective actions for deviations from plans and, if necessary,

(4) modify plans.

Management Control

The following are the **five basic characteristics of an effective control**: (a) Appropriate, (b) Strategic, (c) Acceptable, (d) Reliable and objective, and (e) Cost-effective.

The primary responsibility for exercising control rests with every manager, charged with execution plans.

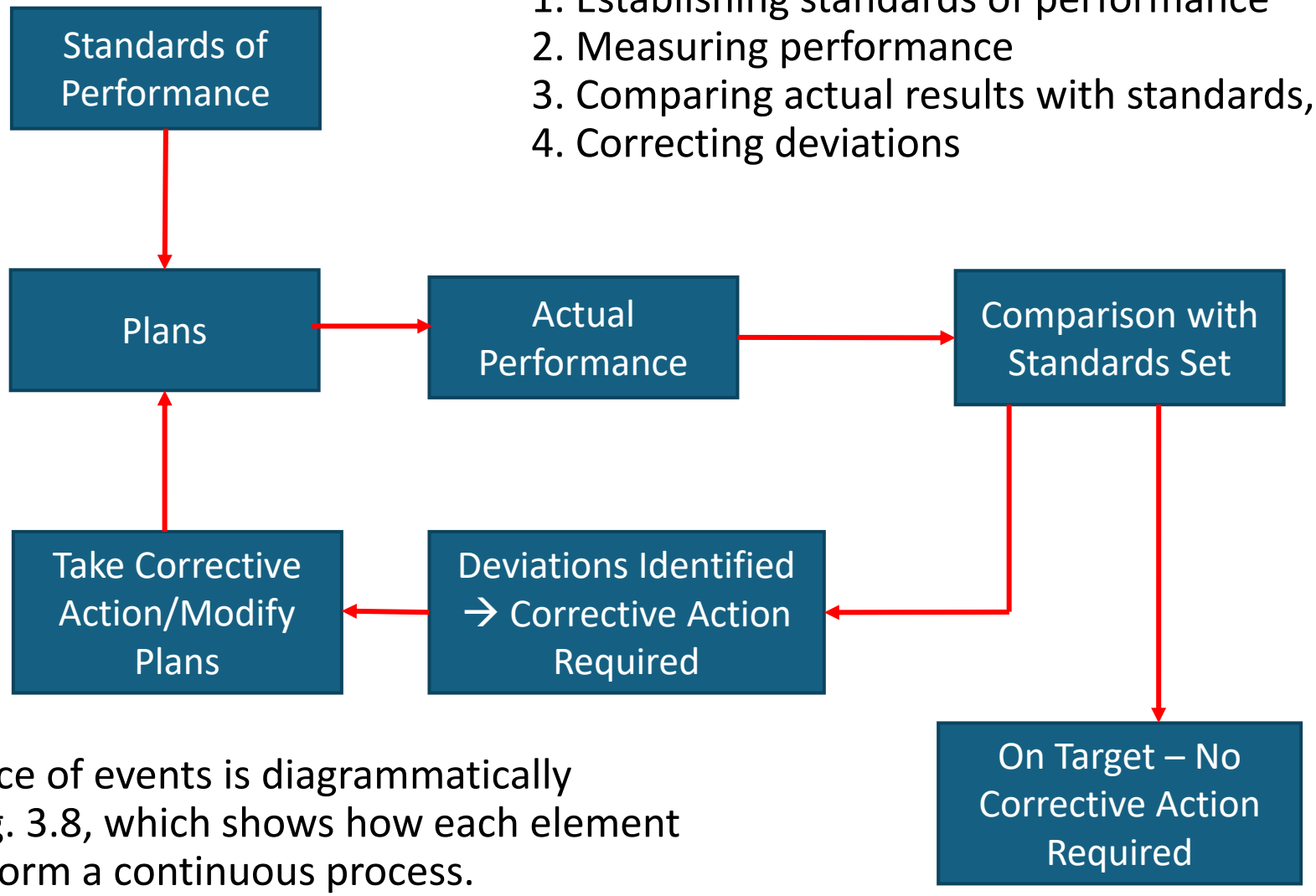
As Henry Fayol said, “In an undertaking, control consists in verifying whether everything occurs in conformity with the plans adopted, the instructions issued and principles established.”

It operates on things, people, actions.” The major prerequisites of control are a plan and a structure. ***The more clear and complete the plan is, the more effective the control can be: plans become the standards by which the actions are measured.***

Control Process

The Control process involves four basic elements:

1. Establishing standards of performance
2. Measuring performance
3. Comparing actual results with standards, and
4. Correcting deviations



This sequence of events is diagrammatically shown in Fig. 3.8, which shows how each element is linked to form a continuous process.

Control Sequence

- First, **the standards of performance need to be verifiable and clearly stated.** e.g. in units of production or in sales volumes. In case standards are qualitative rather than quantitative, it is preferable for them to be expressed in terms of end-results rather than in terms of methods.
- Second, the **measurement of performance** depends heavily on the relevance, adequacy and timely availability of information. **The supply of such information comes from a variety of sources within the organization.** The single-most important source is the management accounting department, which is responsible for the regular production of operating statements, expenditure analyses, profit forecasts, cash flow statements and other relevant control information.
- Third, when comparing actual results with performance, most organizations require action to be taken only when the deviation against standards is significant. Otherwise, no action is taken and no reference upwards is asked for. **This is sometimes called the 'management by exception' principle.**
- Fourth, control is not just a matter of identifying progress, it is also a matter of putting right what may have gone wrong. Hence, **the importance of directing part of the control process to the implementation of appropriate corrective action.**

The information generated by control systems is known as feedback. Feedback is usually produced on results, i.e., on the outputs of the system. Actual performance is recorded and the information fed back to the managers responsible for achieving the target performance. Early feedback is essential for accurate control, especially where unexpected deviations have occurred. Where deviations occur, feedback may indicate the need for a change in the process or its inputs or possibly, a change in the basic plans or original standards.

Feedback in the Control Process

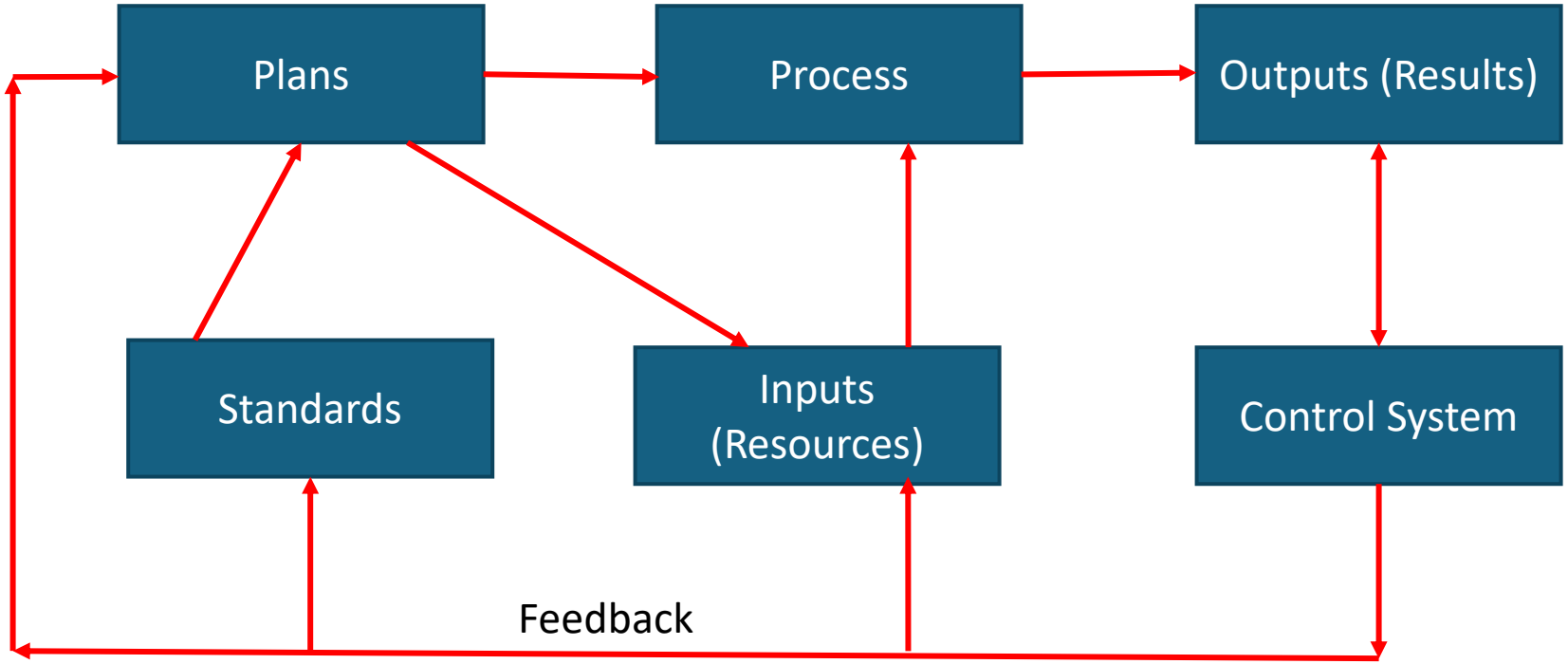


Fig: 3.9 Feedback in the Control System

This sequence of events is diagrammatically shown in Fig. 3.8, which shows how each element is linked to form a ***continuous process ending either in the achievement of targets or in the modification of plans as a result of feedback (Fig. 3.9)***

Control Sequence

- First, **the standards of performance need to be verifiable and clearly stated.** e.g. in units of production or in sales volumes.
  - In case standards are qualitative rather than quantitative, it is preferable for them to be expressed in terms of end results rather than methods.
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Control Sequence and Feedback

The information generated by control systems is known as feedback.

- Feedback is usually produced on results, i.e., on the outputs of the system.
- Actual performance is recorded and the information fed back to the managers responsible for achieving the target performance.
- Early feedback is essential for accurate control, especially where unexpected deviations have occurred.
- Where deviations occur, feedback may indicate the need for a change in the process or its inputs or possibly, a change in the basic plans or original standards.

Systems Management

A system is an orderly arrangement of interdependent activities and related procedures that implement and facilitate the performance of a major activity of an organization.

The basic responsibilities of the systems function:

- reducing clerical costs,
- improving the timeliness of management reports,
- to meet new and more sophisticated management information requirements.
- It concerns itself with the overall complex interrelationship of organization structure, functional divisions of responsibility, and the optimum flow of management information.

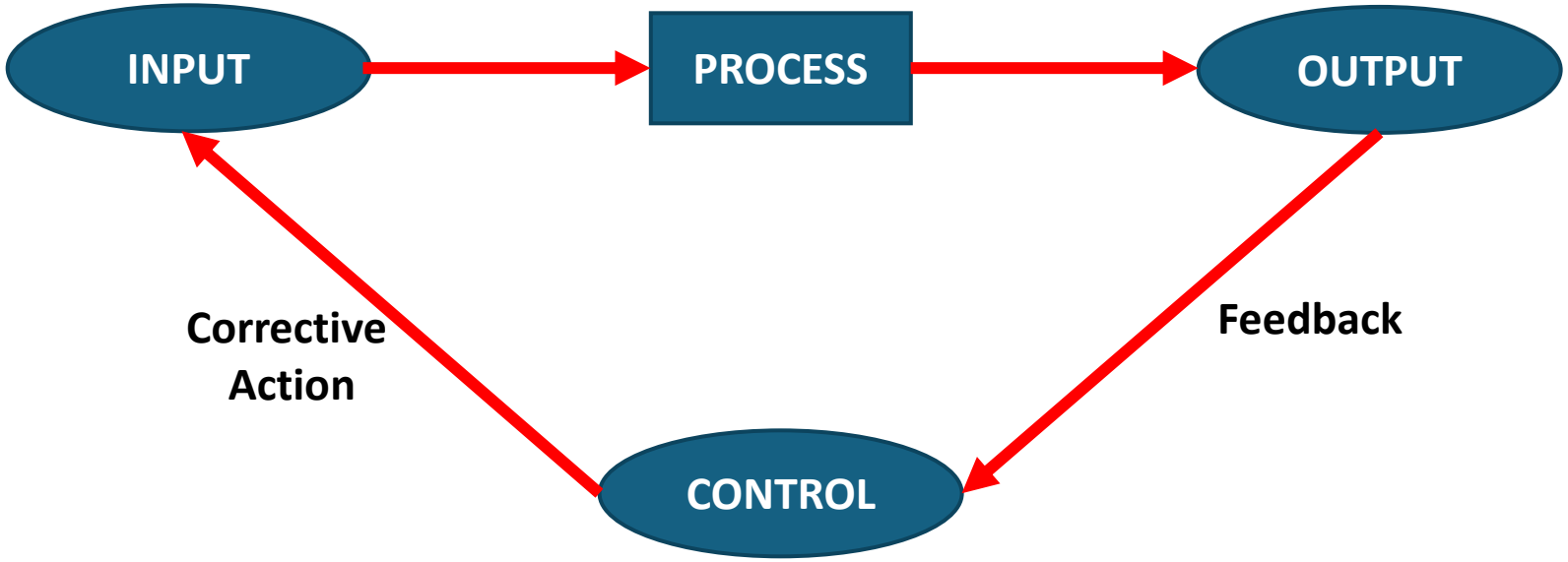
A policy is a basic precept which guides administrative action and defines the authority and the respective relationships required to accomplish the objectives of the organization.

- Policies are more fluid or changeable than objectives since they are subjected to external and internal changes in conditions;
- they relate to all levels of the organization, and are the product of managerial decision making.

A procedure is a series of logical steps by which all repetitive business action is initiated, performed, controlled and finalized.

- A procedure establishes the required action, who is required to act, and when the action is to take place.
- Its essence is a chronological sequence and its implementation is translated into results or actions.

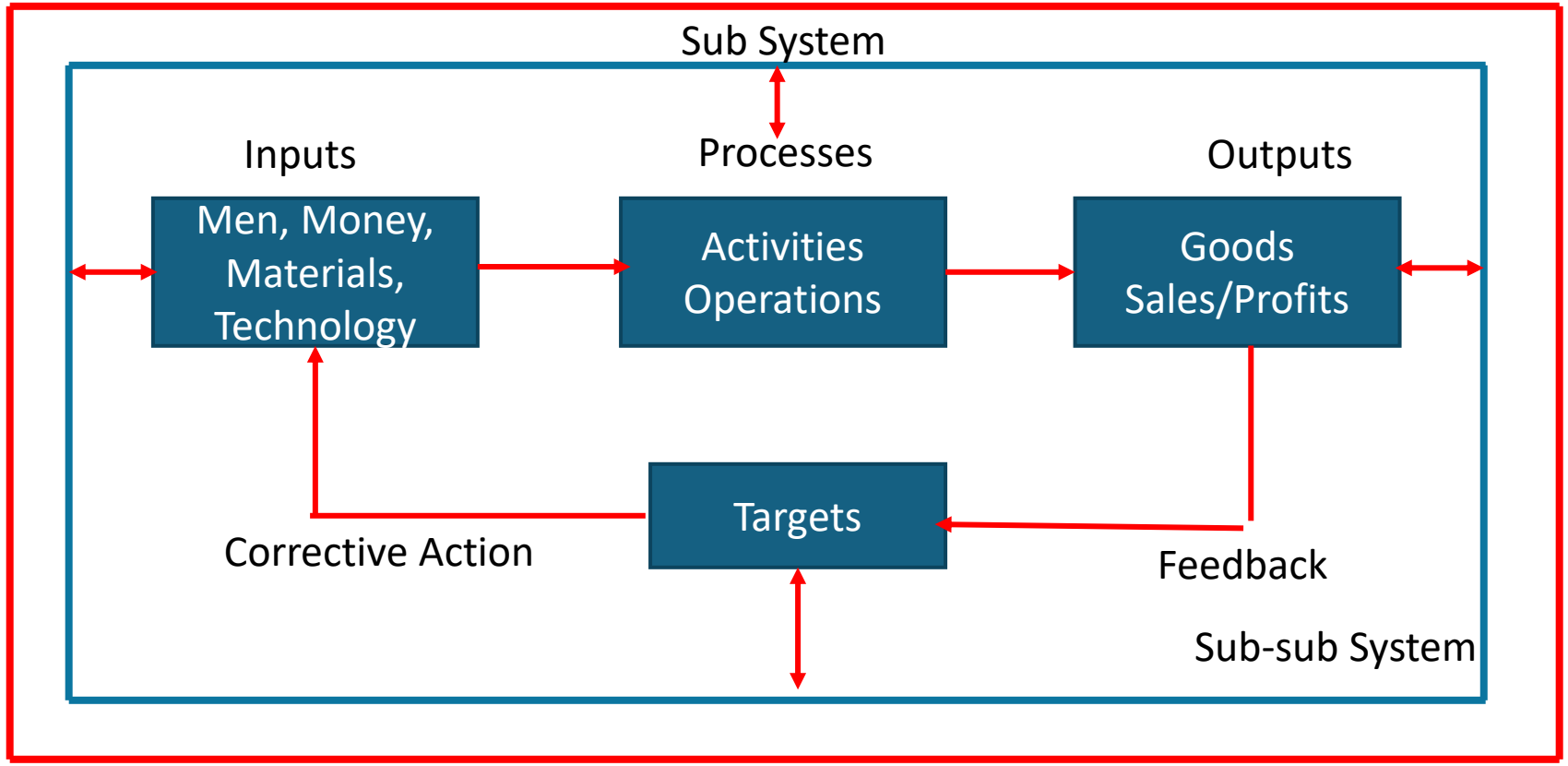
Closed System



Closed System (fig: 3.5 - page 41/42)

Closed System:
A system can be closed or open. A closed system is self-sufficient and self-regulatory and has no interaction with the environment in which it exists (Fig. 3.5). The feedback from the output triggers a control mechanism which then regulates the input to bring back the output to the desired level.

Open System



Open System (fig: 3.6 – page 42)

Open System:
An open system interacts with the environment in which it exists. Figure 3.6 shows an open system. The organization here acts as an open system along with its subsystems, which are the various divisions and departments. At the same time, the organization itself is a subsystem of the environmental system within which it operates.

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PROJECT MANAGEMENT & FINANCE

Human Resource Planning and Management

Lecture 5

Human Resource Management (HRM)

HRM has a Strategic dimension and involves the total deployment of all the human resources available to the firm (Guest, 1987; Legge, 1989).

- HRM is concerned with the wider implications of the management of change – it seeks proactively to encourage flexible attitude and the acceptance of new methods.
- Aspects of HRM constitutes major inputs into organizational development exercises
- HRM is prescriptive and concerned with strategies, the initiation of new activities and the development of fresh ideas.
- HRM determines policies for employment relationships within the enterprise (culture)
- HRM has long-term perspectives, seeking to integrate

HRM Approach

Emphasizes the needs:

- For direct communication with employees rather than their collective representation
- To develop an organizational culture conducive to the adoption of flexible working methods
- For group working and employee participation in group decisions
- To enhance employees' long-term capabilities, not just their competence at current duties

HR Planning Process

- Is the comparison of an organization's existing labor resources with forecast labor demand, and hence scheduling of activities for acquiring, training, redeploying and possibly discarding labor.
- It seeks to ensure – adequate supply of labor is available precisely when required.

Importance:

- Obtain and retain the quantity and quality of human resources it needs
- Makes the best use of its human resources
- Is able to anticipate the problems arising from potential surpluses or deficits of human resource
- Planning of HR should be based on both social and economic goals such as:
- Greater and more creative contribution to productivity
- Need for more broadly skilled managers at the top level
- A stable executive succession
- Need to plan and assimilate changes in status, work, and relationships of employees

HR Planning Process

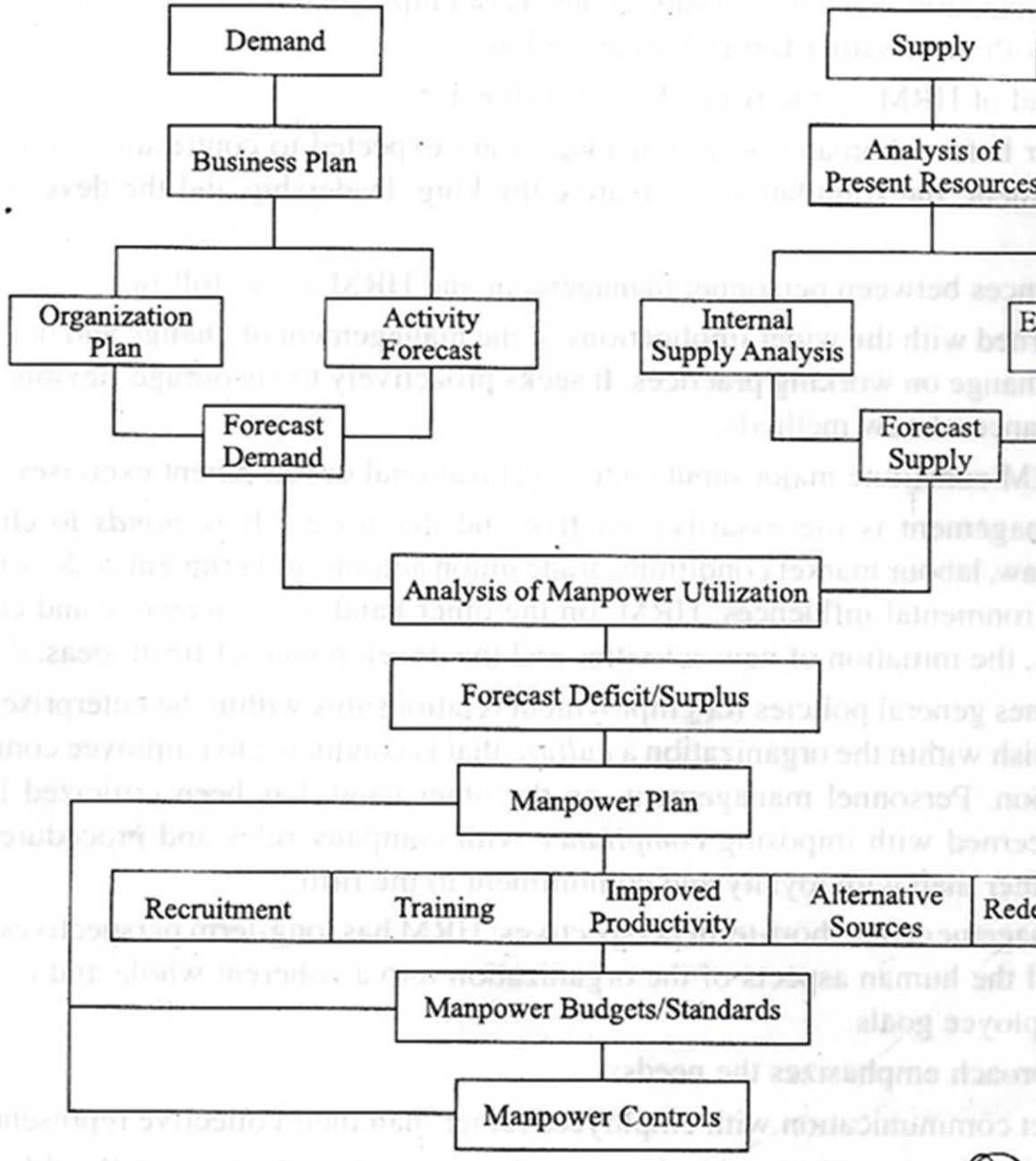

Fig. 4.1 The process of human resource planning

Process of HR Planning

Based on Demand & Supply that leads to Analysis of Manpower Utilization (Page 62 – Fig 4.1)

1. **Demand forecasting**
   -- Estimating the future needs of HR by reference to corporate and functional plans
   -- Forecast of future activity levels
2. **Supply forecasting**
   --Estimating the supply of HR by reference to analysis of current resources and future availability, after allowing wastage
3. **Determining HR requirements**
   -- Analyzing the demand and supply forecasts to identify future deficits or surpluses
4. **Productivity and Cost Analysis**
   -- Analyzing HR productivity, capacity utilization and costs
   -- To identify the need for improvements in productivity or reductions in cost
5. **Action Planning**
   -- Preparing plans to deal with forecast deficits or surpluses of HR
   -- To improve utilization and productivity or reduce HR costs
6. **HR Budgeting and Control**
   --Setting HR budgets and standards
   -- Monitoring the implementation of HR plans against them

Database for Manpower Demand Forecasting

1. **Macro Level** – Comparable past data spanning several years
    a. Population Statistics
    b. Data on Economic Parameters, and
    c. Information on Technologies

2**. Micro Level** – Well defined manpower system is needed for micro forecasting at the enterprise level. The system may have the following modules:
    a. Personal Data Module
    b. Recruitment Module
    c. Job Experience Module
    d. Performance Appraisal Module
    e. Training and Development Module
    f. Miscellaneous Module

Database for Manpower Supply Forecasting

1. Macro level

- Age at entry and at exit
- Annual enrollment and turnover
- Attrition rate
- Retirement age
- Migration
- Morality
- Labor force participation rate

2**. Micro Level** – Well defined manpower system is needed for micro forecasting at the enterprise level. The system may have the following modules:

a. Personal Data Module
b. Recruitment Module
c. Job Experience Module
d. Performance Appraisal Module
e. Training and Development Module
f. Miscellaneous Module

Productivity

- Represents the output of goods and services which can be obtained from a given input of employees.
- Productivity should be monitored by using such measures as

i. Manpower costs per unit of output
ii. Manpower costs as a ratio of sales value
iii. Sales value per employee
iv. Tons of product handled per man-hour
v. labor cost as a percentage of added value (the difference between production cost and sales value)

Manpower Cost

Manpower Cost
- can be grouped under the following headings:

I. Remuneration Costs
II. Recruitment Costs
III. Training Costs
IV. Relocation Costs
V. Leaving Costs
VI. Support Costs
VII. Personnel Administration Costs

Action Planning

- The manpower planning should be prepared on the basis of analysis of

(a) Manpower requirements and

(b) A study of the implications of the information on the productivity and costs.

I. The recruitment plan
II. The re-development plan
III. The redundancy plan
IV. The training plan
V. The productivity plan
VI. The retention plan

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PROJECT MANAGEMENT & FINANCE

Human Resource Planning and Management

Lecture 6

Recruitment and Selection

Recruitment is the process of identifying the prospective employees, stimulating and encouraging them to apply for a particular job in an organization.

Policy formulating is an essential managerial function. It is a complicated process which may involve all levels of employees. While formulating a policy, values and objectives of different groups involved should be kept in view so as to raise efficiency of working relationships in an organization.

In addition, the recruitment policy shall have to be in line with the manpower requirements in order to attain the objectives of the organization.

The recruitment process involves five different elements:

(a) a recruitment policy;

(b) a recruiting organization;

(c) a forecast of manpower requirements;

(d) development of manpower sources; and

(e) techniques for utilizing these sources.

Recruitment and Selection - Stages

The overall aim of the recruitment and selection process should be to obtain at minimum cost the number and quality of employees required to satisfy the manpower needs of the company.

There are three stages of recruitment and selection:

(a) Defining requirements: Preparing job descriptions and specifications; deciding terms and conditions of employment.

(b) Attracting candidates: Reviewing and evaluating alternative sources of applicants, inside and outside the company; advertising; using agencies and consultants.

(c) Selecting candidates: Sifting through applications, interviewing, testing, assessing candidates; offering employment, obtaining references; preparing terms of reference of employment.

Recruitment - Policy

The recruitment policy states the objectives and provides a framework for implementation of the recruitment program in the form of procedures.

- A recruitment policy may include issues such as promotion within, or transfer from, the organization (as quotas are usually fixed in government and public sector organizations as also quotas for certain minority groups).
- While the recruitment system is a function related to the personnel function, in principle, recruitment should remain a line responsibility.
- There is a need to constantly review and improve the methods of recruitment in view of the changing times and demand.
- A sound recruitment program necessarily involves appraisal of each source, and the techniques, from the viewpoint of relative quality of personnel which it has provided.
- The valuation procedures should consist of existing employees in terms of their jobs—success, evaluation of source from which good and poor employees were recruited and the methods used in assessing the relative values.

Recruitment and Selection Policy- Objectives

Some of the important objectives of the recruitment policy are:

(a) Providing maximum employment security to individual employees and avoiding frequent lay-off or lost time;

(b) Assuring fairness to all employees in all employment relationships, including promotions and transfers;

(c) Encouraging each employee of the organization to continue development of his talents and skills;

(d) Following the guidelines of the relevant public policy on hiring and employment relationship;

(e) Safeguarding the interests of minorities and handicapped in the recruitment of personnel;

(f) Assuring employees interest in their personal goals and employment objectives;

Recruitment - Methods

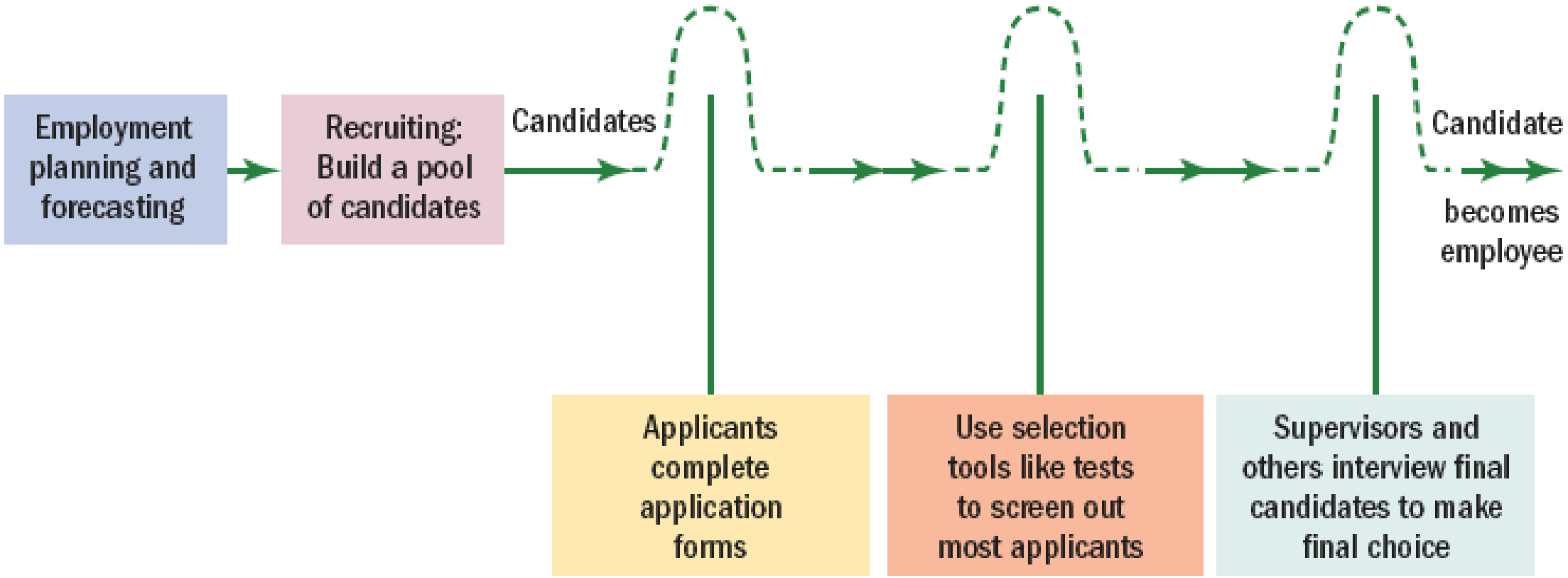
Methods of Recruitment Broadly, methods of recruitment can be divided into three categories: (a) Direct methods; (b) Indirect methods; and (c)Third-party methods.

Direct methods: These methods include sending recruiters to educational and professional institutions.

Indirect methods: These methods cover advertising in newspapers, trade and professional journals, technical journals, and on television.

Third-party methods: These include commercial and private employment agencies, state or public employment agencies, placement offices of schools, colleges and professional associations, recruiting firms, management consulting firms, indoctrination seminars for college professors, trade unions, temporary help agencies, casual labor sources, and the process of deputation

Steps in the Recruitment and Selection Process



The recruitment and selection process is a series of hurdles aimed at selecting the best candidate for the job.

Recruitment and Selection Process

Recruitment is the process of identifying the prospective employees, stimulating and encouraging them to apply for a particular job in an organization.

1. Decide what positions you'll have to fill through personnel planning and forecasting.
2. Build a pool of candidates for these jobs by recruiting internal or external candidates.
3. Have candidates complete application forms and perhaps undergo an initial screening interview.
4. Use selection techniques like tests, background investigations, and physical exams to identify viable candidates.
5. Decide who to make an offer to, by having the supervisor and perhaps others on the team interview the candidates.

Selection Process & Tests

The selection process involves the following seven steps:

(a) Preliminary screening of applications,

(b) Review of application forms,

(c) Physical examination,

(d) Checking references,

(e) Psychological testing,

(f) Interview, and

(g) Internal promotion program.

Selection tests can be put into four categories:

(a) Achievement or intelligence tests,

(b) Aptitude or potential ability tests,

(c) Personality tests, and

(d) Interest tests

Induction and Placement

The induction and placement program simply proposes that new recruits shall be made to feel and develop themselves as a part of the organization as early as possible.

It primarily consists of three steps:

(a) General orientation by the staff: The employee is given general information about the history and operations of the firm. The purpose is to help an employee build up some interest in the organization;

(b) Specific orientation by the job supervisor: The employee is shown the department and his place of work, the location of facilities and is told about specific practices and customs of the organization. The aim is to enable the employee to adjust to his work and environment; and

(c) Follow-up orientation: The purpose is to find out whether the employee is reasonably well satisfied with the organization. Personal talks, guidance and counselling efforts are made to remove the difficulties experienced by the new entrant.

A crucially important aspect of the induction process is that of informing recruits where to go for help if they experience problems. Entrants should know whom to approach, and the correct procedures to follow. Appropriate contacts might be supervisors, personnel officers, higher managers or trade union representatives.

The “BIG FIVE” – Characteristics Employers are looking for

- Extraversion
  - The tendency to be sociable, assertive, active, and to experience positive effects, such as energy and zeal.
- Emotional stability/neuroticism
  - The tendency to exhibit poor emotional adjustment and experience negative effects, such as anxiety, insecurity, and hostility.
- Openness to experience
  - The disposition to be imaginative, nonconforming, unconventional, and autonomous.
- Agreeableness
  - The tendency to be trusting, compliant, caring, and gentle.
- Conscientiousness
  - Is comprised of two related facets: achievement and dependability.

Basic Features of Interview

- An interview
  - A procedure designed to obtain information from a person through oral responses to oral inquiries
- Types of interviews
  - Selection interview
  - Appraisal interview
  - Exit interview
- Interviews formats
  - Structured
  - Unstructured

Types of Interview

- Selection interview
  - A selection procedure designed to predict future job performance on the basis of applicants' oral responses to oral inquiries.
- Appraisal interview
  - A discussion, following a performance appraisal, in which supervisor and employee discuss the employee's rating and possible remedial actions.
- Exit interview
  - An interview to elicit information about the job or related matters to the employer some insight into what's right or wrong about the firm.

Formats of Interview

- Unstructured or nondirective interview
  - An unstructured conversational-style interview in which the interviewer pursues points of interest as they come up in response to questions.
  - An interview in which each interviewer forms an independent opinion after asking different questions.
- Structured or directive interview
  - An interview following a set sequence of questions.
  - An interview in which the applicant is interviewed sequentially by several persons; each rates the applicant on a standard form.

Interview Contents – Types of Questions

- Situational Questions
  - A series of job-related questions that focus on how the candidate would behave in a given situation.
- Behavioral Questions
  - A series of job-related questions that focus on how they reacted to actual situations in the past.
- Job-related Questions
  - A series of job-related questions that focus on relevant past job-related behaviors.

Sample Questions

Situational Questions:

- 1. Suppose a co-worker was not following standard work procedures. The co-worker was more experienced than you and claimed the new procedure was better. Would you use the new procedure?
- 2. Suppose you were giving a sales presentation and a difficult technical question arose that you could not answer. What would you do?

Past Behavior Questions:

- 3. Based on your past work experience, what is the most significant action you have ever taken to help out a co-worker?
- 4. Can you provide an example of a specific instance where you developed a sales presentation that was highly effective?

Background Questions:

- 5. What work experiences, training, or other qualifications do you have for working in a teamwork environment?
- 6. What experience have you had with direct point-of-purchase sales?

Job Knowledge Questions:

- 7. What steps would you follow to conduct a brainstorming session with a group of employees on safety?
- 8. What factors should you consider when developing a television advertising campaign?

Six (06) Factors Affecting Interview

- First impressions
  - The tendency for interviewers to jump to conclusions—make snap judgments—about candidates during the first few minutes of the interview.
  - Negative bias: unfavorable information about an applicant influences interviewers more than does positive information.
- Misunderstanding the job
  - Not knowing precisely what the job entails and what sort of candidate is best suited causes interviewers to make decisions based on incorrect stereotypes of what a good applicant is.
- Candidate-order error
  - An error of judgment on the part of the interviewer due to interviewing one or more very good or very bad candidates just before the interview in question.
- Nonverbal behavior and impression management
  - Interviewers' inferences of the interviewee's personality from the way he or she acts in the interview have a large impact on the interviewer's rating of the interviewee.
  - Clever interviewees attempt to manage the impression they present to persuade interviewers to view them more favorably.

Six (06) Factors Affecting Interview

- Effect of personal characteristics: attractiveness, gender, race
  - Interviewers tend have a less favorable view of candidates who are:
    - Physically unattractive
    - Female
    - Of a different racial background
    - Disabled
- Interviewer behaviors affecting interview outcomes
  - Inadvertently telegraphing expected answers.
  - Talking so much that applicants have no time to answer questions.
  - Letting the applicant dominate the interview.
  - Acting more positively toward a favored (or similar to the interviewer) applicant.

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PROJECT MANAGEMENT & FINANCE

Human Resource Planning and Management

Lecture 7

Training & Development

Training and development programs are necessary in any organization to improve the quality of work of the employees at all levels, particularly in a world of fast changing technology, changing values, and deteriorating environment. **The purpose of both programs is similar**. **The main difference between the two** concerns the levels of employees for whom these programs are meant, and the contents and techniques employed.

Training is a short-term process utilizing a systematic and organized procedure by which non managerial personnel acquire technical knowledge and skills for a definite purpose. It refers to instructions in technical and mechanical operations such as operation of a machine.

Development is a long-term educational process utilizing a systematic and organized procedure by which managerial personnel get conceptual and theoretical knowledge. It involves broader education and its purpose is long-term development.

Training & Benefits

Training can be defined

- as the modification of behavior through experience;
- the transfer of skills and knowledge from those who have them to those who do not; or
- to bring about significant improvements in performance as a result of instructions, practice and experience.

These definitions indicate what training is but not as to how it can be made effective. The only way of achieving good results from training is to tackle it systematically.

Benefits:

- Reduction in waste and spoilage.
- Improvement in methods of work.
- Reduction in learning time.
- Reduction in supervisory burden.
- Reduction in machine breakdowns and maintenance costs.
- Reduction in accident rate.
- Improvement in quality of products.
- Improvement in production rate.
- Improvement in morale and reduction in grievances.
- Improvement in efficiency and productivity.
- Personal Growth

Training Objectives

Following are the three basic objectives of training:

(a) Preparing employees for the job meant for them while on first appointment or transfer, or on promotion, and imparting them the required skill and knowledge;

(b) Assisting employees to function more effectively by exposing them to the latest concepts, information, techniques, and developing the skills required to perform in their respective areas; and

(c) Building a second line of competent officers and preparing them to occupy more responsible positions

Management Development

Management development is a systematic process of training and growth by which managerial personnel gain and supply skills, knowledge, attitudes and the insight into managing the work in their organizations effectively and efficiently.

Need for Management Development Management development is necessary for the following reasons:

1. Society is facing a rapid technological and social change. Management have to tackle problems arising out of introduction of automation, intense market competition, growth of new markets, enlarged labor participation in management and more interest taken by the public and the government in various activities of business.
2. Business and industrial leaders are increasingly recognizing their social and public responsibilities which call for a much broader outlook on the part of management.
3. Management-labor relations are becoming increasingly complex.
4. For handling problems arising out of increasing size and complexity of the organization.
5. For understanding and adjusting to changes in socio-economic forces, including changes in public policy and concepts of social justice, industrial democracy, problems of ecology (smog or pollution), ekistics (problems of human settlements), ergonomics (problems of working environment) and cultural anthropology (problems of fitting machines to men).

Objectives – Top Management

1. To improve thought processes and analytical ability in order to uncover and examine problems and take decisions in the best interests of the organization and the country.
2. To broaden the outlook of the executive towards his role, position, and responsibilities within the organization and outside.
3. To think through problems which may confront the organization now or in the future.
4. To understand economic, technical and institutional forces in order to solve business problems.
5. To acquire knowledge about the complexity of human relations

Objectives – Middle Management

1. To establish a clear picture of executive functions and responsibilities.
2. To bring about an awareness of the broad aspects of management problems, and an appreciation of interdepartmental relations.
3. To develop an ability to analyse problems and take appropriate action.
4. To develop familiarity with the managerial uses of financial accounting, psychology, business law and business statistics.
5. To inculcate knowledge of human motivation and human relationships.
6. To develop responsible leadership.

Objectives – Functional (Lower-level) Management

1. To increase knowledge of business functions and operations in specific fields in marketing, production, finance, personnel.
2. To increase proficiency in management techniques such as work study, inventory control, operations research, quality control.
3. To stimulate creative thinking in order to improve methods and procedures.
4. To understand the functions performed in a company.
5. To understand human relations.
6. To develop the ability to analyse problems in one’s area or functions.

Motivation

Motivation must work as a major factor in the study and practice of management.

Organized effort— that is, getting work done through and in association with others—requires motivating interdependent effort, as opposed to effort that is self-cancelling.

Thus, Henri Fayol, one of the early observers of organized effort, shares the contemporary opinion that motivation constitutes the core of management.

'Motivation Human Resource Planning and Management refers to the degree of readiness of an organism to pursue some designated goal and implies the determination of the nature and locus of the forces inducing the degree of readiness.'

Motivation will occur only under specific conditions referred to by three key notions (Victor Vroom):

1. expectancy,
2. instrumentality, and
3. valence,

Thus, individuals will be motivated to the degrees that:

1. They expect that their effort will lead to enhanced performance.
2. They believe that enhanced performance will be instrumental in leading to their reward, both extrinsic and intrinsic.
3. They value the expected reward which, in terminology of the expectancy theory, has a high valence for individuals.

Motivation of Employees in Actual Practice

According to FW Taylor, “It is for the employer to find the right way that could make workers more productive.” He writes that an employee is a bundle of untapped energy, ready to work if he is properly trained and fairly treated.

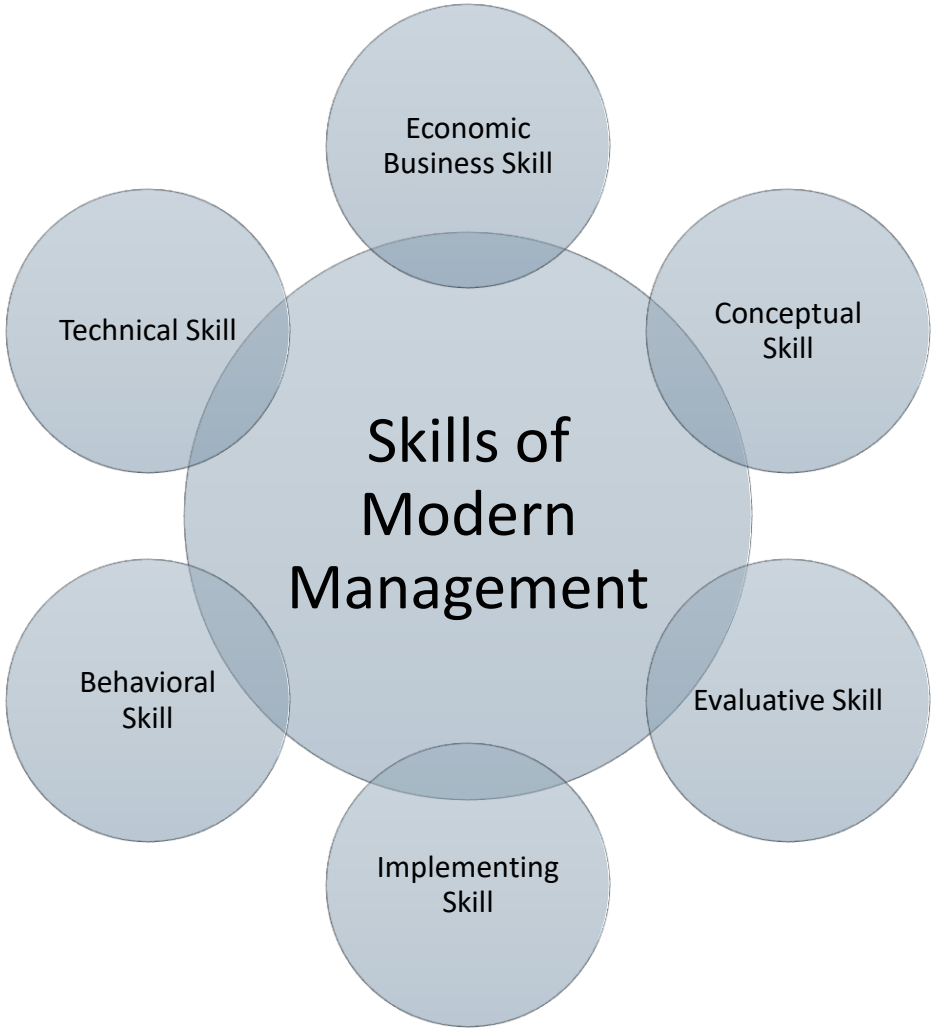
In managing an organization, some of the important factors which could improve motivation and morale of employees are:

1. A well-defined humanistic personnel policy.
2. A long-term employment policy.
3. Fair wages, timely salary administration and reward for good performance.
4. Performance appraisal and merit rating.
5. Handling of workers’ grievances in a cordial environment.
6. Handling disciplinary cases conforming to the principles of natural justice.
7. Participative management, supervision and increased autonomy in work.
8. Job enrichment and growth.
9. Self-motivation by top management, senior and middle level managers, and supervisors.

Managerial Skills

Skill is the ability to apply knowledge effectively and readily in the performance of a physical/mental task.

- Knowledge does not make the skill.
- Skill is the ingenuity with which knowledge is used to guide thought, analysis, and judgments in responding to conditional needs in a variety of complex situations.
- Managers exercise this ability with varying degrees of competency and proficiency.
- Knowledge is defined here as the “stockpile” of valid information and facts organized from past experiences with specific problems.



Technical Skills

This skill requires the proficient use of the specialized information, methods, processes, procedures, and techniques involved in the technology of the product or services of the company or industry.

Examples include:

- Ability to sense and forecast trends in the growth and development of the organization’s technology.
- Ability to communicate and understand technical terms of organizational specialists.
- Ability to exploit new knowledge to innovate services or products.

Behavioral Skills

This skill is primarily the utilization of knowledge to understand people as they conduct themselves. Such an understanding promotes working with others cooperatively and effectively as a natural and continuous activity. It requires an awareness of attitudes and beliefs held by individuals and groups and how these govern their goal-reaching processes.

Examples include:

- Ability to sense people's needs and drives, and relate them to planned accomplishments
- Ability to coordinate various power groups
- Ability to empathize with various lifestyles.

Conceptual Skills

This skill is the ability to utilize the existing knowledge in order to acquire additional knowledge. It is purposeful risk-taking for creating new conditions. Exploring, questioning, and probing are some of the tools that cut across established areas of a business. Conceptual skill is interdisciplinary.

Examples include:

- Ability to generate alternatives from present and past experiences.
- Ability to see the whole from the given constituent parts.
- Ability to predict ends from certain beginnings even without an orientation.

Implementing Skills

This skill involves the ability to get work done on a day-to-day, program-to-program basis, or the ability to concentrate just enough resources of time, money, and effort to initiate an activity, conduct the activity, and bring it to a successful end. Achievement and accomplishments are the principal dimensions for employing this skill.

Examples include:

- Ability to sense and make use of pace, sequence, and time.
- Ability to work on a task to completion with intensified drive.
- Ability to foresee barriers and pursue circumventing steps.

Business-economic Skills

This skill is the utilization of knowledge to understand a business enterprise operating in a market for the sale of goods and services. It involves the manner in which an enterprise uses scarce and limited resources to meet demands. It involves the strategy of production, distribution, and consumption of goods and services with max output and min of input.

Examples include:

- Ability to move ahead, as a member of an organization, towards coordinated ends.
- Ability to undertake the pursuit of a gain in the marketplace in the face of risk and uncertainty.
- Ability to compete for positive results in the free enterprise system.

Evaluative Skills

This skill is the ability to appraise the value or the degree of worth of a process or accomplishment. This skill tends to be quantitative since its basic idea is “how much”.

Examples include:

- Ability to analyze a complex mass of information into numeric and quantities.
- Ability to measure deviation, variation, and drift from prescribed directions.
- Ability to initiate feedback corrections to reduce variance within prescribed times.